(20)

In St2:- $x+3$ is +ve when x values as either $-1,-2,-3,$
$0,1,2,3 \ldots \infty$. means x values $-1,-2,-3,0,1,2,3 \ldots \infty$
Means we can't say yes/no, because yes is for some assumptions & no is for some other assumptions.
We shouldn't consider this case as sufficient one.
Answer is St1 → Option [A]

Ex:-6:- If 'n' is an integer is $(n+1)$ odd?
St1:- $n+2$ is an even integer.
St2:- $n-1$ is an odd integer
Ans:- from statement 1, to match '$n+2$' is an even the suitable 'n' values are all even integers from $-\infty$ to $+\infty$
$n = -\infty \ldots -8, -6, -4, -2, 0, 2, 4, 6, 8, \ldots \infty$
then $n+1 = -8+1 = -7, -6+1 = -5 \ldots 4+1 = 5, 8+1 = 9 =$ odd
So, answer yes to 1st statement
It sufficient alone
from St2:- to satisfy $(n-1)$ is odd suitable n values are again $n = -\infty, \ldots -8, -6, -4, -2, 0, 2, 4, 6, 8, \ldots \infty$.
here also answer for $(n+1)$ is odd. So, it's also sufficient alone. Answer yes to this.
From the above analysis.
Both statements alone sufficient to answer the question. Answer option is [D]

Ex:-7:- What is the weight of each identical pole?

St1:- Weight of each identical pole of 20% is 40.

St2:- Weight of 3 identical poles is 100 kgs greater than weight of 2 identical poles.

Ans:- from St1:- $20\%\ W = 40 \Rightarrow 10\% = 20$

$\Rightarrow 100\%$ of $W = 200$ kgs

Sufficient to find weight alone.

from St2:- $3W = 2W + 100 \Rightarrow W = 100$ kgs

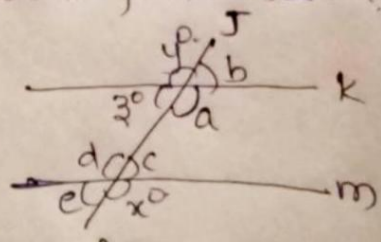
It is also sufficient alone but $W = 100$ kgs.

Note:- When two statements alone sufficient to answer the question means on both statements the answer must be same, otherwise comes under Contradiction and choose it as not sufficient to both statements.

Answer: Option [E]

Ex:-8:- In the figure, If lines k & m are ||el, what is $\angle x = ?$

St1:- $y = 120$, St2:- $z = 60$



When two ||el st lines are intersected by 3rd st line, then this is called as transverse st line & properties are like corresponding, suplimentary angles are equal.

St1:- $x° = d° = y° = 120$ Sufficient.

St2:- $z° = e° = 60° \Rightarrow e° + x° = 180° \Rightarrow x = 120°$ sufficient

Answer: Option [D]